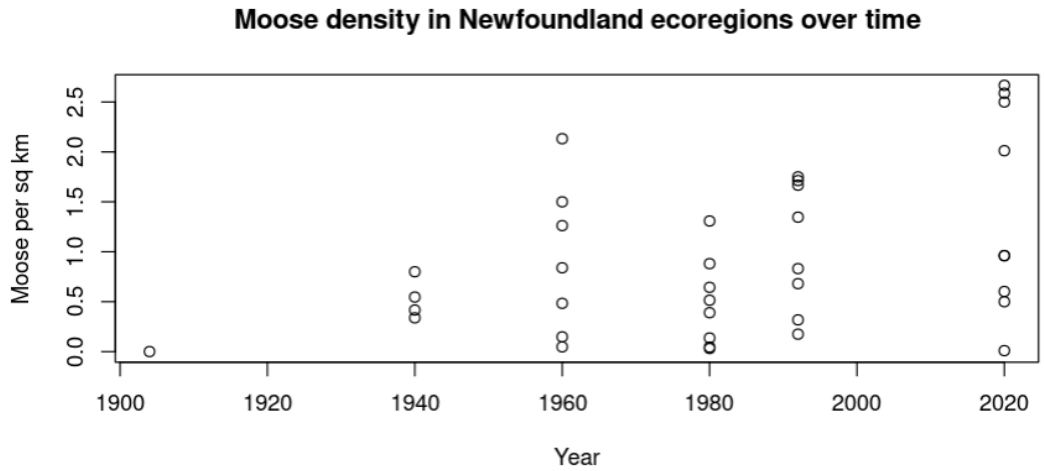
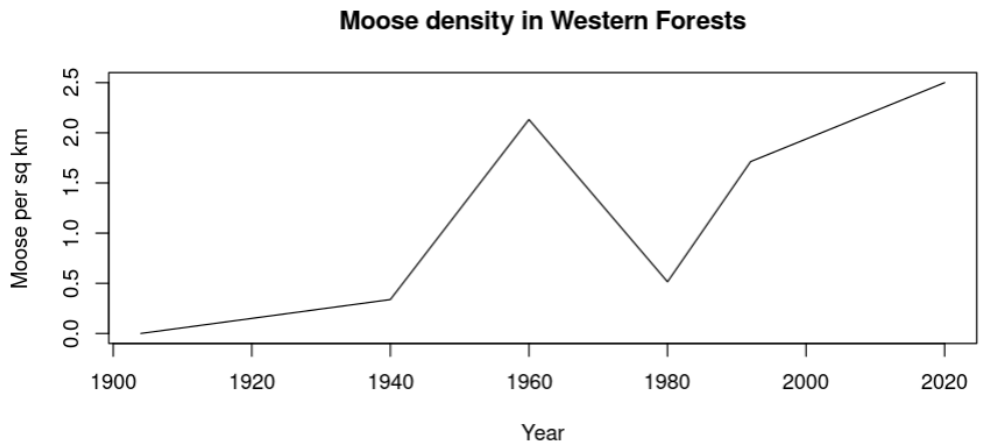





```
> # QUESTION 1
> # Installing dplyr
> install.packages("dplyr")
> # Loading dplyr
> library(dplyr)
> # QUESTION 2
> moosedata <- read.csv("MoosePopulation (2) .csv")
> # QUESTION 3
> moose_clean <- na.omit(moosedata)
                 Ecoregion Year  Area
1        North_Shore_Forests 2020  8400
2  Northern_Peninsula_Forests 2020 12000
3         Long_Range_Barrens 2020  8800
4            Central_Forests 2020 20500
5            Western_Forests 2020  6800
6  EasternHyperOceanicBarrens 2020  4400
7           Maritime_Barrens 2020 21300
8             Avalon_Forests 2020  4200
9   StraitOfBelleIsleBarrens 2020   500
```

10	North_Shore_Forests	1992	8400
11	Northern_Peninsula_Forests	1992	12000
12	Long_Range_Barrens	1992	8800
13	Central_Forests	1992	20500
14	Western_Forests	1992	6800
15	EasternHyperOceanicBarrens	1992	4400
16	Maritime_Barrens	1992	21300
17	Avalon_Forests	1992	6400
19	North_Shore_Forests	1980	8400
20	Northern_Peninsula_Forests	1980	12000
21	Long_Range_Barrens	1980	8800
22	Central_Forests	1980	20500
23	Western_Forests	1980	6800
24	EasternHyperOceanicBarrens	1980	4400
25	Maritime_Barrens	1980	21300
26	Avalon_Forests	1980	6400
28	North_Shore_Forests	1960	8400
29	Northern_Peninsula_Forests	1960	12000
30	Long_Range_Barrens	1960	8800
31	Central_Forests	1960	20500
32	Western_Forests	1960	6800
34	Maritime_Barrens	1960	21300
35	Avalon_Forests	1960	4200
37	North_Shore_Forests	1940	8400
38	Northern_Peninsula_Forests	1940	12000
40	Central_Forests	1940	20500
41	Western_Forests	1940	6800
50	Western_Forests	1904	6800

	Estimated_Moose_Pop	Observation_Method
1	21740	Aerial_Survey
2	32000	Aerial_Survey
3	5300	Camera_Trap
4	41250	Aerial_Survey
5	17000	Expert_Estimate
6	2200	Expert_Estimate
7	20450	Expert_Estimate
8	4040	Aerial_Survey
9	5	Camera_Trap
10	14000	Aerial_Survey
11	21000	Aerial_Survey
12	1540	Expert_Estimate
13	27600	Aerial_Survey
14	11650	Aerial_Survey
15	1400	Camera_Trap

16	14500	Aerial_Survey
17	5320	Expert_Estimate
19	7400	Aerial_Survey
20	15700	Expert_Estimate
21	400	Camera_Trap
22	13200	Aerial_Survey
23	3500	Aerial_Survey
24	600	Expert_Estimate
25	8300	Hunter_Reports
26	200	Camera_Trap
28	10600	Aerial_Survey
29	18000	Aerial_Survey
30	1300	Hunter_Reports
31	17210	Aerial_Survey
32	14500	Expert_Estimate
34	10300	Aerial_Survey
35	200	Expert_Estimate
37	3500	Expert_Estimate
38	9600	Expert_Estimate
40	11200	Expert_Estimate
41	2300	Expert_Estimate
50	4	Count

	Data_Source
1	GovOfNewfoundland
2	NGO_Monitoring
3	MemorialUniversity
4	GovOfNewfoundland
5	ParksCanada
6	MemorialUniversity
7	MemorialUniversity
8	MemorialUniversity
9	ParksCanada
10	GovOfNewfoundland
11	NGO_Monitoring
12	MemorialUniversity
13	GovOfNewfoundland
14	GovOfNewfoundland
15	ParksCanada
16	GovOfNewfoundland
17	MemorialUniversity
19	GovOfNewfoundland
20	MemorialUniversity
21	ParksCanada
22	GovOfNewfoundland

```
23  GovOfNewfoundland
24 MemorialUniversity
25  GovOfNewfoundland
26        ParksCanada
28  GovOfNewfoundland
29  GovOfNewfoundland
30  GovOfNewfoundland
31  GovOfNewfoundland
32        ParksCanada
34  GovOfNewfoundland
35        ParksCanada
37  GovOfNewfoundland
38  GovOfNewfoundland
40        ParksCanada
41        ParksCanada
50  GovOfNewfoundland
> moose_clean <- na.omit(moosedata)
> View(moose_clean)
> # QUESTION 4
> select(moose_clean, Ecoregion, Year, Area, Estimated_Moose_Pop)
1         North_Shore_Forests 2020  8400
2  Northern_Peninsula_Forests 2020 12000
3         Long_Range_Barrens 2020  8800
4            Central_Forests 2020 20500
5            Western_Forests 2020  6800
6  EasternHyperOceanicBarrens 2020  4400
7           Maritime_Barrens 2020 21300
8             Avalon_Forests 2020  4200
9   StraitOfBelleIsleBarrens 2020   500
10        North_Shore_Forests 1992  8400
11 Northern_Peninsula_Forests 1992 12000
12         Long_Range_Barrens 1992  8800
13            Central_Forests 1992 20500
14            Western_Forests 1992  6800
15 EasternHyperOceanicBarrens 1992  4400
16           Maritime_Barrens 1992 21300
17             Avalon_Forests 1992  6400
19        North_Shore_Forests 1980  8400
20 Northern_Peninsula_Forests 1980 12000
21         Long_Range_Barrens 1980  8800
22            Central_Forests 1980 20500
23            Western_Forests 1980  6800
24 EasternHyperOceanicBarrens 1980  4400
25           Maritime_Barrens 1980 21300
```

26	Avalon_Forests	1980	6400
28	North_Shore_Forests	1960	8400
29	Northern_Peninsula_Forests	1960	12000
30	Long_Range_Barrens	1960	8800
31	Central_Forests	1960	20500
32	Western_Forests	1960	6800
34	Maritime_Barrens	1960	21300
35	Avalon_Forests	1960	4200
37	North_Shore_Forests	1940	8400
38	Northern_Peninsula_Forests	1940	12000
40	Central_Forests	1940	20500
41	Western_Forests	1940	6800
50	Western_Forests	1904	6800

Estimated_Moose_Pop	
1	21740
2	32000
3	5300
4	41250
5	17000
6	2200
7	20450
8	4040
9	5
10	14000
11	21000
12	1540
13	27600
14	11650
15	1400
16	14500
17	5320
19	7400
20	15700
21	400
22	13200
23	3500
24	600
25	8300
26	200
28	10600
29	18000
30	1300
31	17210
32	14500

```
34              10300
35                200
37               3500
38               9600
40              11200
41               2300
50                  4
> moose_sel <- select(moose_clean, Ecoregion, Year, Area, Estimated_Moose_Pop)
> # QUESTION 5 (a&b)
> # Observations in dataset
> min(moose_sel$Year)
[1] 1904
> year_min <- min(moose_sel$Year)
> max(moose_sel$Estimated_Moose_Pop)
[1] 41250
> moose_max <- max(moose_sel$Estimated_Moose_Pop)
> # QUESTION 6
> # Calculating moose density
> mutate(moose_sel, MooseDensity = Estimated_Moose_Pop / Area)
> moosedata2 <- mutate(moose_sel, MooseDensity = Estimated_Moose_Pop / Area)
                 Ecoregion Year  Area
1        North_Shore_Forests 2020  8400
2  Northern_Peninsula_Forests 2020 12000
3         Long_Range_Barrens 2020  8800
4            Central_Forests 2020 20500
5            Western_Forests 2020  6800
6  EasternHyperOceanicBarrens 2020  4400
7           Maritime_Barrens 2020 21300
8            Avalon_Forests 2020  4200
9   StraitOfBelleIsleBarrens 2020   500
10       North_Shore_Forests 1992  8400
11 Northern_Peninsula_Forests 1992 12000
12        Long_Range_Barrens 1992  8800
13           Central_Forests 1992 20500
14           Western_Forests 1992  6800
15 EasternHyperOceanicBarrens 1992  4400
16          Maritime_Barrens 1992 21300
17            Avalon_Forests 1992  6400
19       North_Shore_Forests 1980  8400
20 Northern_Peninsula_Forests 1980 12000
21        Long_Range_Barrens 1980  8800
22           Central_Forests 1980 20500
23           Western_Forests 1980  6800
24 EasternHyperOceanicBarrens 1980  4400
```

25	Maritime_Barrens	1980	21300
26	Avalon_Forests	1980	6400
28	North_Shore_Forests	1960	8400
29	Northern_Peninsula_Forests	1960	12000
30	Long_Range_Barrens	1960	8800
31	Central_Forests	1960	20500
32	Western_Forests	1960	6800
34	Maritime_Barrens	1960	21300
35	Avalon_Forests	1960	4200
37	North_Shore_Forests	1940	8400
38	Northern_Peninsula_Forests	1940	12000
40	Central_Forests	1940	20500
41	Western_Forests	1940	6800
50	Western_Forests	1904	6800

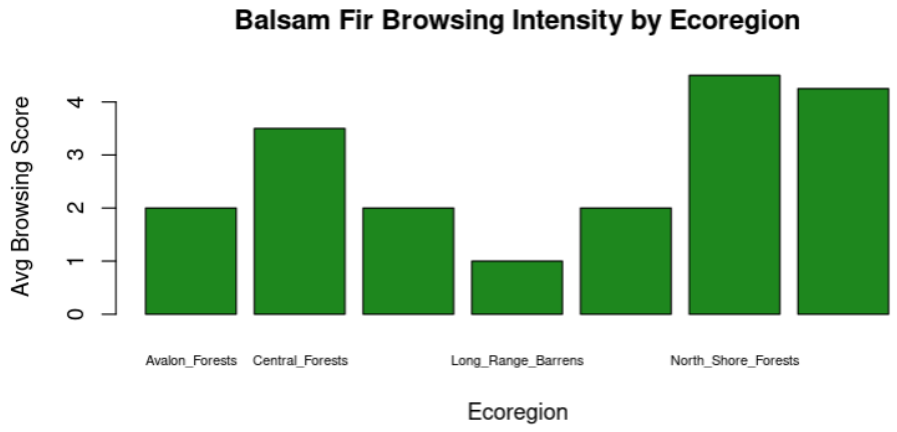
	Estimated_Moose_Pop	MooseDensity
1	21740	2.5880952381
2	32000	2.6666666667
3	5300	0.6022727273
4	41250	2.0121951220
5	17000	2.5000000000
6	2200	0.5000000000
7	20450	0.9600938967
8	4040	0.9619047619
9	5	0.0100000000
10	14000	1.6666666667
11	21000	1.7500000000
12	1540	0.1750000000
13	27600	1.3463414634
14	11650	1.7132352941
15	1400	0.3181818182
16	14500	0.6807511737
17	5320	0.8312500000
19	7400	0.8809523810
20	15700	1.3083333333
21	400	0.0454545455
22	13200	0.6439024390
23	3500	0.5147058824
24	600	0.1363636364
25	8300	0.3896713615
26	200	0.0312500000
28	10600	1.2619047619
29	18000	1.5000000000
30	1300	0.1477272727
31	17210	0.8395121951

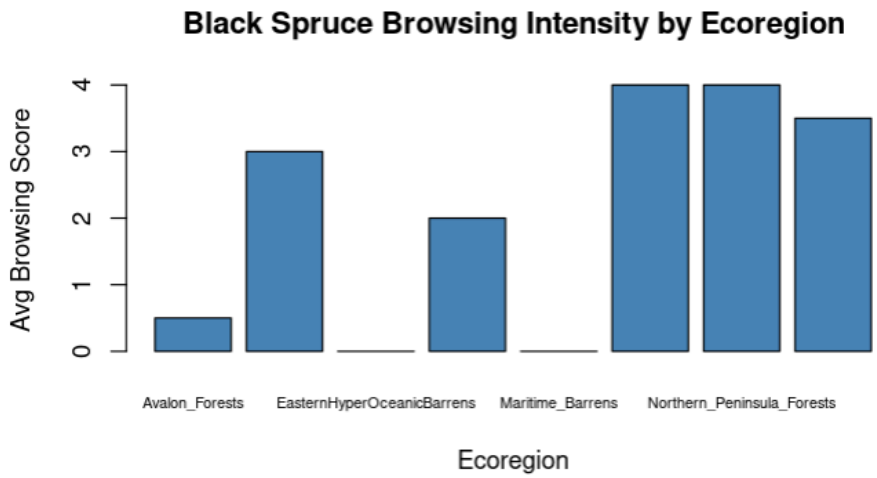
```
32              14500 2.1323529412
34              10300 0.4835680751
35                200 0.0476190476
37               3500 0.4166666667
38               9600 0.8000000000
40              11200 0.5463414634
41               2300 0.3382352941
50                  4 0.0005882353
> # QUESTION 7
> plot(moosedata2$Year, moosedata2$MooseDensity,
  +     xlab = "year",
  +     ylab = "Moose per sq km",
  +     main = "Moose density in Newfoundland ecoregions over time")
> # QUESTION 8
> # Western forests data
> filter(moosedata2, Ecoregion == "Western_Forests")
> moose_west <- filter(moosedata2, Ecoregion == "Western_Forests")
> # Showing change in moose density------
> plot(moose_west$Year, moose_west$MooseDensity,
      +     type = "l",
      +     xlab = "Year",
      +     ylab = "Moose per sq km",
      +     main = "Moose density in Western Forests over time")
> # QUESTION 9
> # Saving 2020 dataset
> filter(moosedata2, Year == 2020)
> moose_2020 <- filter(moosedata2, Year == 2020)
> # High densities in 2020 dataset
> filter(moose_2020, MooseDensity > 2.0)
> moose_2020_high <- filter(moose_2020, MooseDensity > 2.0)
> arrange(moose_2020_high, desc(MooseDensity))
> moose_2020_high_byD <- arrange(moose_2020_high, desc(MooseDensity))
> # QUESTION 10

> # QUESTION 20
> # distribution
> # No the dataset is not evenly distributed, as the two had different numbers of individuals
sampled. Balsam_Fir is overrepresented (11 samples), while Black_Ash is underrepresented
with only 1 sample.
> # It is important to recognize bias in datasets because small sample sizes can lead to
misleading averages. For example, if the only Black_Ash tree sampled happened to be heavily
browsed, it makes the entire species look like a preferred food source, even if that isn't true for
the whole population.
> # QUESTION 21
```

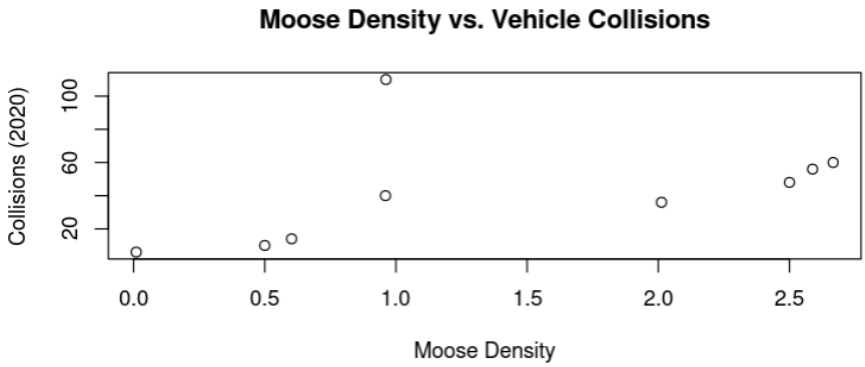
```
>moose_sap <- left_join(moose_2020, sap_clean, by = 'Ecoregion', relationship =
"many-to-many")
> moose_2020b <- moose_clean %>%
  +     filter(Year == 2020) %>%
  +     mutate(MooseDensity = Estimated_Moose_Pop / Area)
> moose_sap <- left_join(moose_2020b, sap_clean, by = 'Ecoregion', relationship =
"many-to-many")
> # QUESTION 22
> sum_spe_browse <- moose_sap %>%
  +     group_by(Species, Ecoregion) %>%
  +     summarize(AvgBrowsing = mean(BrowsingScore),
  +               AvgDensity = mean(MooseDensity)) %>%
  +     print()
# A tibble: 38 × 4
# Groups:   Species [6]
Species      Ecoregion AvgBrowsing AvgDensity
<chr>        <chr>           <dbl>      <dbl>
1 "Alder "     Avalon_F…         4        0.962
2 "Alder "     Central_…         4.5      2.01
3 "Alder "     EasternH…         4        0.5
4 "Alder "     Long_Ran…         3.5      0.602
5 "Alder "     Maritime…         3.5      0.960
6 "Alder "     North_Sh…         4.5      2.59
7 "Alder "     Northern…         5        2.67
8 "Alder "     Western_…         5        2.5
9 "Balsam_Fir" Avalon_F…         2        0.962
10 "Balsam_Fir" Central_…         3.5      2.01
# i 28 more rows
# i Use `print(n = ...)` to see more rows
> # QUESTION 23
> #Hypothesis Evidence: Yes, there is evidence. At lower densities, browsing scores vary wildly
between species (selective), while at higher densities, the scores for all species tend to cluster
higher and more closely together (generalist).
> #Preferences: Balsam Fir typically shows the highest browsing scores (most favored), while
Black Spruce often shows the lowest (least favored).
> #Missing Species: Black Ash is often missing from these figures because it had a sample size
of only 1, making it impossible to calculate a meaningful average for density correlation.
> # QUESTION 24
> collisions2020 <- c(56, 60, 14, 36, 48, 10, 40, 110, 6)
> human_pop <- c(18000, 12000, 4000, 75100, 24000, 3500, 32000, 270000, 2300)
> study_sites <- c("North_Shore_Forests","Northern_Peninsula_Forests",
"Long_Range_Barrens","Central_Forests","Western_Forests","EasternHyperOceanicBarrens","
Maritime_Barrens","Avalon_Forests","StraitOfBelleIsleBarrens")
moose_coll <- data.frame(collisions2020, human_pop, study_sites)
```

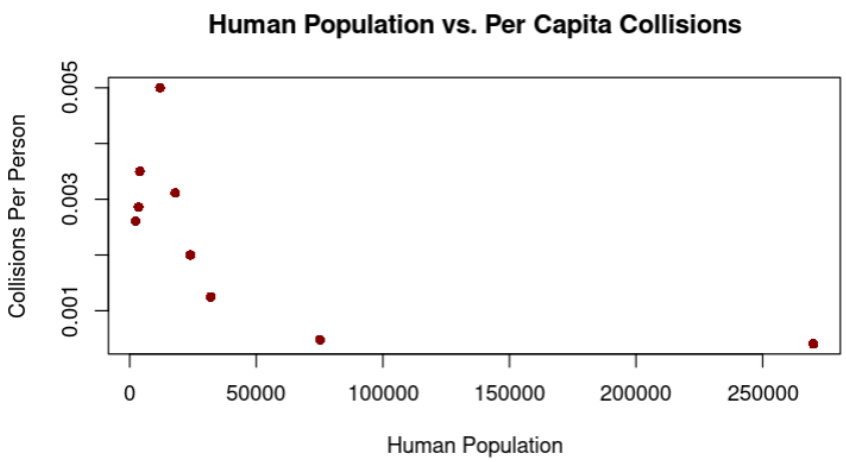
```
> # QUESTION 25
> moose_coll2 <- moose_coll %>%
  +     rename(Ecoregion = study_sites)
> coll_merge <- left_join(moose_coll2, moose_2020, by = "Ecoregion")
> # QUESTION 26
> plot(coll_merge$MooseDensity, coll_merge$collisions2020,
        +     xlab = "Moose Density", ylab = "Collisions (2020)",
        +     main = "Moose Density vs. Vehicle Collisions")
```

```
> # QUESTION 26
> plot(coll_merge$MooseDensity, coll_merge$collisions2020,
        +     xlab = "Moose Density", ylab = "Collisions (2020)",
        +     main = "Moose Density vs. Vehicle Collisions")
> # Generally, higher moose density correlates with more collisions. But, yes there is an outlier
where the moose density is 1, and the # of collisions is over 100. Other than that as moose
density increases collisions increase.
> # QUESTION 27
> coll_merge_per_capita <- coll_merge %>%
  +     mutate(coll_per_capita = collisions2020 / human_pop)
> # QUESTION 28
> plot(coll_merge_per_capita$human_pop, coll_merge_per_capita$coll_per_capita,
      +     xlab = "Human Population",
      +     ylab = "Collisions Per Person",
      +     main = "Human Population vs. Per Capita Collisions",
      +     pch = 16,
      +     col = "darkred")
> # QUESTION 29
> The trend shows that as the human population increases, the number of moose-vehicle
collisions per person actually decreases. This makes sense because rural ecoregions have
smaller human populations but much higher moose densities, meaning each individual person
living there has a statistically higher chance of encountering a moose on the road.
```


Balsam Fir Browsing Intensity by Ecoregion
Avg Browsing Score
0
1
2
3
4
Avalon_Forests
Central_Forests
Long_Range_Barrens
North_Shore_Forests
Ecoregion


Black Spruce Browsing Intensity by Ecoregion
Avg Browsing Score
0
1
2
3
4
Avalon_Forests
EasternHyperOceanicBarrens
Maritime_Barrens
Northern_Peninsula_Forests
Ecoregion


Moose Density vs. Vehicle Collisions
Collisions (2020)
20
60
100
0.0
0.5
1.0
1.5
2.0
2.5
Moose Density


Human Population vs. Per Capita Collisions
Collisions Per Person
0.001
0.003
0.005
0
50000
100000
150000
200000
250000
Human Population